

CodSoft AI Internship - Task 1 Report

Task Name: Chatbot with Rule-Based Responses | Domain: Artificial Intelligence

1. Objective & Problem Overview

Design and implement a rule-based conversational chatbot in Python. The system processes natural language inputs, normalizes text strings, and matches predefined keyword patterns to deliver contextually relevant responses or fallbacks.

2. System Architecture & Methodology

The system utilizes Python's regular expressions ('re' module) for intent classification and pattern matching. User inputs are normalized to lowercase and matched against regular expression boundaries. The chatbot includes multiple randomized response variations using the 'random' module to provide dynamic conversation flow.

3. Python Source Code Implementation

```
import re
import random

responses = {
    "greeting": ["Hello! How can I assist you?", "Hi there!", "Hey! Ready to code?"],
    "how_are_you": ["Operating at 100% efficiency!", "I am doing great!"],
    "thanks": ["You're welcome!", "Happy to help!"],
    "fallback": ["I don't have a rule for that yet.", "Could you rephrase?"]
}

def codsoft_bot(user_input):
    user_input = user_input.lower()
    if re.search(r'\b(hello|hi|hey)\b', user_input):
        return random.choice(responses["greeting"])
    elif re.search(r'\b(how are you|how is it going)\b', user_input):
        return random.choice(responses["how_are_you"])
    elif re.search(r'\b(thank you|thanks)\b', user_input):
        return random.choice(responses["thanks"])
    elif re.search(r'\b(rules|requirements)\b', user_input):
        return "Complete at least 3 tasks, maintain GitHub repo CODSOFT_TASKSNO, and post on LinkedIn."
    else:
        return random.choice(responses["fallback"])

while True:
    text = input("You: ")
    if re.search(r'\b(quit|exit)\b', text.lower()):
        break
    print("Bot:", codsoft_bot(text))
```

4. Sample Execution & Test Output

```
User: Hello
Bot: Hi there! Ready to code?

User: What are the rules?
Bot: Complete at least 3 tasks, maintain GitHub repo CODSOFT_TASKSNO, and post on LinkedIn.

User: quit
Bot: Good luck with your CodSoft internship tasks! Goodbye.
```